VIT-AP UNIVERSITY

Fundamentals of Electrical and Electronics Engineering (FEEE)

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Lecture-19

Faculty

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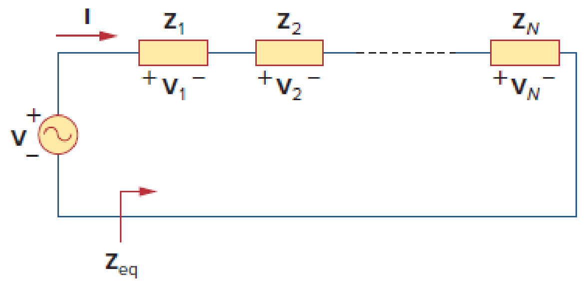
VII. Kirchhoff's Laws in the Frequency Domain:

- We cannot do circuit analysis in the frequency domain without Kirchhoff's current and voltage laws. Therefore, we need to express them in the frequency domain.

- For KVL, let $v1, v2, \ldots, vn$ be the voltages around a closed loop. Then

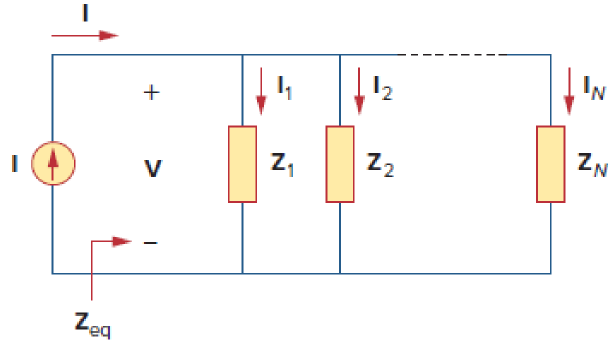


$$v1 + v2 + \cdots + vn = 0$$

$$V_{m1}\cos(\omega t + \vartheta_1) + V_{m2}\cos(\omega t + \vartheta_2) + \cdots + V_{mn}\cos(\omega t + \vartheta_n) = 0$$

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0$$

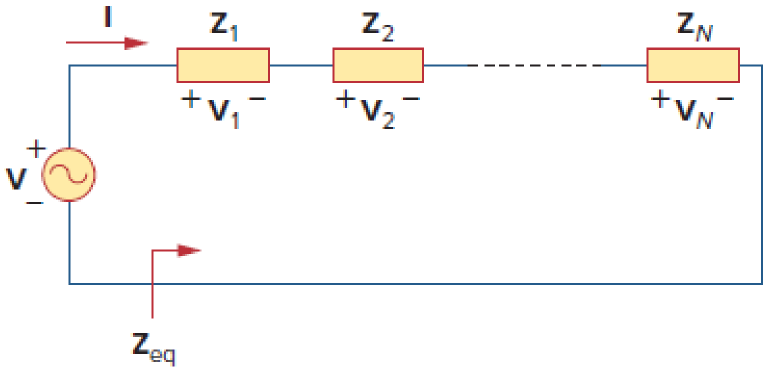
[indicating that Kirchhoff's voltage law holds for phasors.]



- If $\mathbf{I}_1, \mathbf{I}_2, \ldots, \mathbf{I}_n$ are the phasor forms of the sinusoids $i_1, i_2, \ldots, i_n$, then

$$\mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_n = 0$$

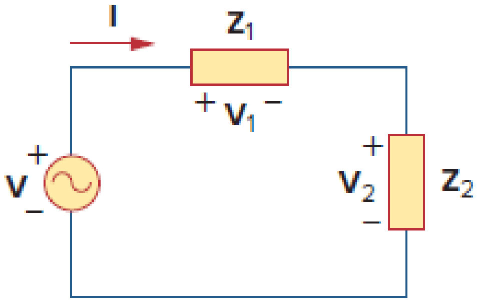
Impedance Combinations



$$\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_N = \mathbf{I}(\mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N)$$

$$\mathbf{Z}_{eq} = \frac{\mathbf{V}}{\mathbf{I}} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N$$

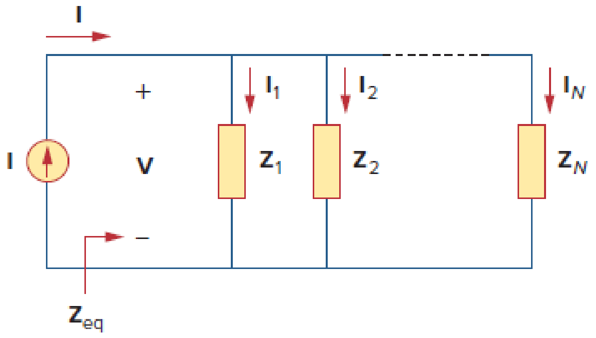
$$\boxed{\mathbf{Z}_{eq} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N}$$



$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}_1 + \mathbf{Z}_2}$$

$\mathbf{V}_1 = \mathbf{Z}_1\mathbf{I}$ and $\mathbf{V}_2 = \mathbf{Z}_2\mathbf{I}$, then

$$\boxed{\mathbf{V}_1 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}, \qquad \mathbf{V}_2 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}}$$

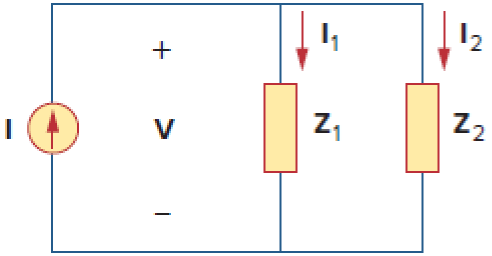


$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_N = \mathbf{V}\left(\frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \cdots + \frac{1}{\mathbf{Z}_N}\right)$$

$$\frac{1}{\mathbf{Z}_{eq}} = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \cdots + \frac{1}{\mathbf{Z}_N}$$

the equivalent admittance is

$$\boxed{\mathbf{Y}_{eq} = \mathbf{Y}_1 + \mathbf{Y}_2 + \cdots + \mathbf{Y}_N}$$



$$\mathbf{Z}_{eq} = \frac{1}{\mathbf{Y}_{eq}} = \frac{1}{\mathbf{Y}_1 + \mathbf{Y}_2} = \frac{1}{1/\mathbf{Z}_1 + 1/\mathbf{Z}_2} = \frac{\mathbf{Z}_1\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$$

$$\mathbf{V} = \mathbf{I}\mathbf{Z}_{eq} = \mathbf{I}_1\mathbf{Z}_1 = \mathbf{I}_2\mathbf{Z}_2$$

the currents in the impedances are

$$\boxed{\mathbf{I}_1 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{I}, \qquad \mathbf{I}_2 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{I}}$$

1.The voltage $v = 12\cos(60t + 45°)$ is applied to a 0.1-H inductor. Find the steady-state current through the inductor.

For the inductor, $\mathbf{V} = j\omega L\mathbf{I}$, where $\omega = 60$ rad/s

$$\mathbf{V} = 12\,\underline{/45^\circ}\text{ V}.$$

$$\mathbf{I} = \frac{\mathbf{V}}{j\omega L}$$

$$= \frac{12\,\underline{/45^\circ}}{j60 \times 0.1} = \frac{12\,\underline{/45^\circ}}{6\,\underline{/90^\circ}} = 2\,\underline{/-45^\circ}\text{ A}$$

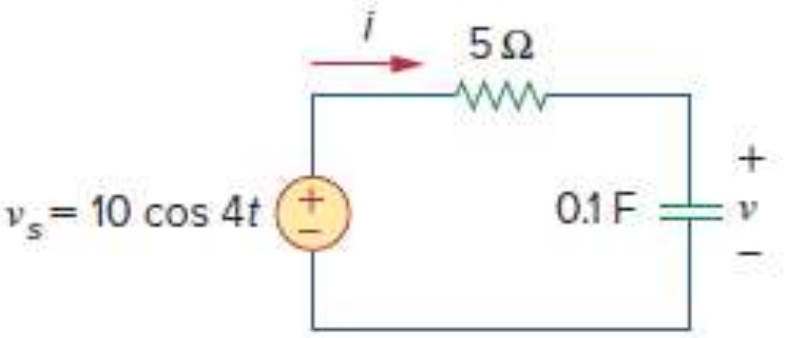
Converting this to the time domain,

$$i(t) = 2\cos(60t - 45^\circ)\text{ A}$$

2. If voltage $v = 25\sin(100t - 15°)$ V is applied to a 50 μF capacitor, calculate the current through the capacitor.

$$\mathbf{I} = j\omega C\mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$

3. Find $v(t)$ and $i(t)$ in the circuit shown in Fig.



From the voltage source $10\cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10\underline{/0^\circ} \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5\ \Omega$$

Hence the current

$$\mathbf{I} = \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10\underline{/0^\circ}}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2}$$
$$= 1.6 + j0.8 = 1.789\underline{/26.57^\circ} \text{ A}$$

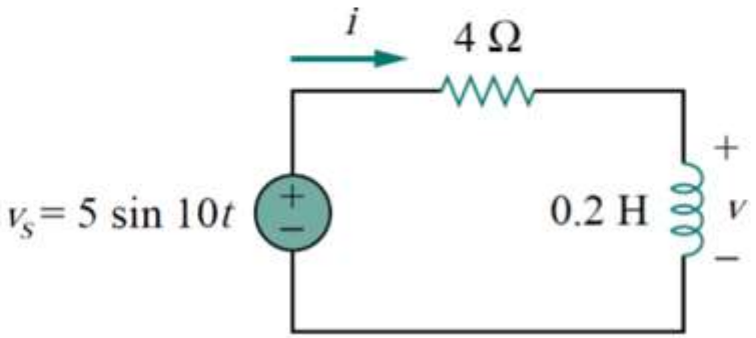
The voltage across the capacitor is

$$\mathbf{V} = \mathbf{I}\mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789\underline{/26.57^\circ}}{j4 \times 0.1}$$
$$= \frac{1.789\underline{/26.57^\circ}}{0.4\underline{/90^\circ}} = 4.47\underline{/-63.43^\circ} \text{ V}$$

$$i(t) = 1.789\cos(4t + 26.57^\circ) \text{ A}$$
$$v(t) = 4.47\cos(4t - 63.43^\circ) \text{ V}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.

4. Determine *v*(*t*) and *i*(*t*).



$$I = \frac{V_s}{Z}$$

$$X_L = 10 \times 0.2 = 2\ \Omega$$

$$Z = 4 + j2\ \Omega$$

$$I = \frac{V_s}{Z} = \frac{5\angle 0^o}{4 + j2} = 1.118\angle 26.57^o\ A$$

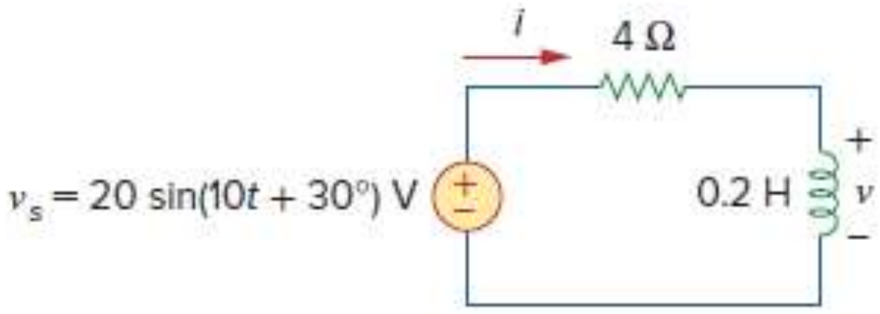
$$V = \left(1.118\angle 26.57^o\right) \times (j2)$$

$$V = 2.236\angle -63.43^o\ V$$

$$i(t) = 1.118\cos\left(10t + 26.57^o\right)\ A$$

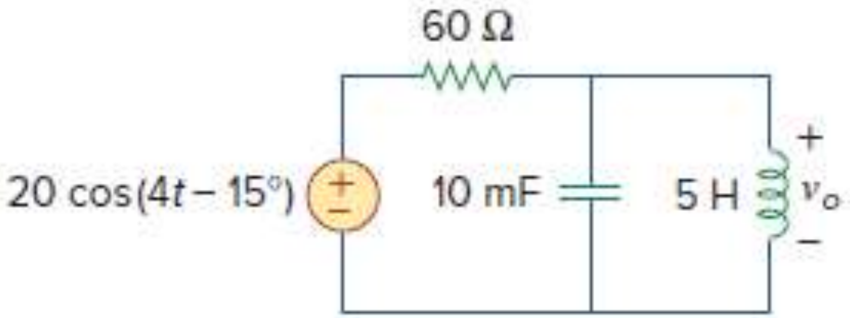
$$v(t) = 2.236\cos\left(10t - 63.43^o\right)\ V$$

5. Determine $v(t)$ and $i(t)$.



6. Determine $v_o(t)$ in the circuit of Fig. below

- To do the analysis in the frequency domain, we must first transform the time domain circuit into the phasor domain equivalent.

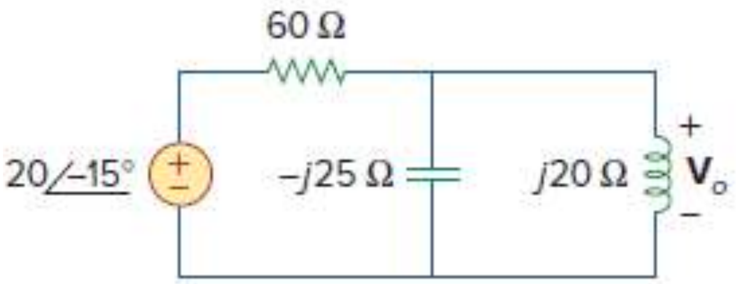


$$v_s = 20\cos(4t - 15^\circ) \quad \Rightarrow \quad \mathbf{V}_s = 20\angle{-15^\circ}\ \text{V},$$

$$10\ \text{mF} \quad \Rightarrow \quad \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}}$$

$$\omega = 4 \qquad = -j25\ \Omega$$

$$5\ \text{H} \quad \Rightarrow \quad j\omega L = j4 \times 5 = j20\ \Omega$$



$\mathbf{Z}_1$ = Impedance of the 60-Ω resistor

$\mathbf{Z}_2$ = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

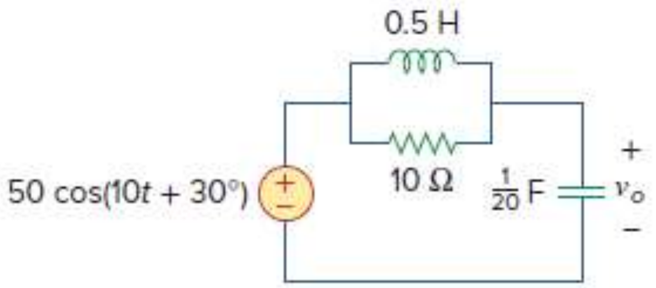
$$\mathbf{Z}_1 = 60\ \Omega \qquad \mathbf{Z}_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100\ \Omega$$

$$\mathbf{V}_o = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}_s = \frac{j100}{60 + j100}(20\angle{-15^\circ}) = (0.8575\angle{30.96^\circ})(20\angle{-15^\circ})$$

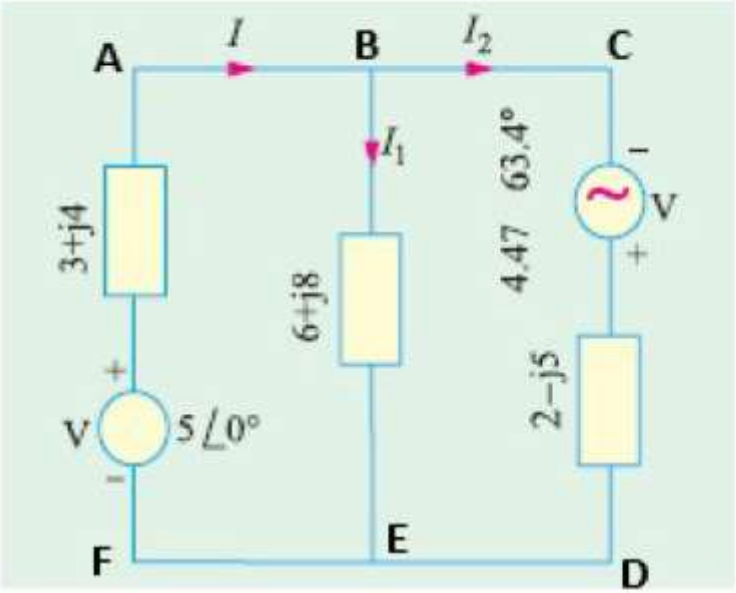
$$= 17.15\angle{15.96^\circ}\ \text{V}$$

$$v_o(t) = 17.15\cos(4t + 15.96^\circ)\ \text{V}$$

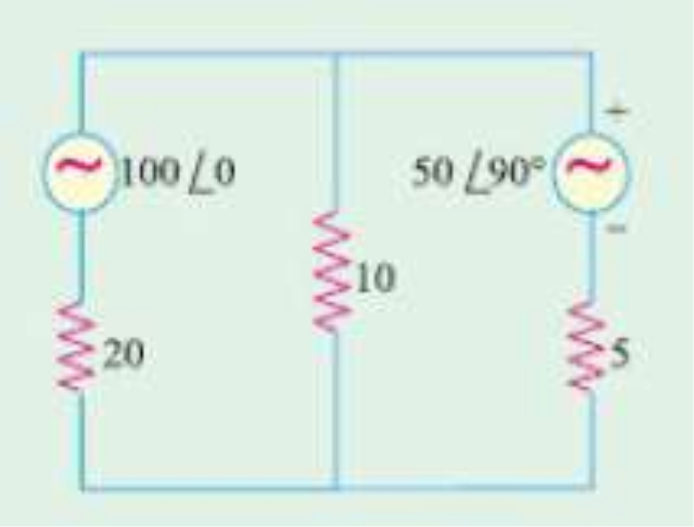
7. Determine $v_o(t)$ in the circuit of Fig. below



8) Using Kirchhoff's Laws, calculate the current flowing through each branch of the circuit shown in figure below.



2)Use Kirchhoff's laws to find the current flowing in each branch of the network shown in figure below.



9.Find current **I** in the circuit of Fig.